

Suvadip Sana

PhD Candidate, Department of Statistics and Data Science, Cornell University

ss2776@cornell.edu · [Personal Website](#) · [Google Scholar](#)

Research Interests

I develop the mathematical and algorithmic foundations for AI systems that must operate under diverse, often conflicting human preferences. My work spans **AI value alignment**, **preference learning**, **democratic AI**, and **behavioral evaluation of language models**, drawing on statistical theory and social choice. Broader interests include **AI safety** and the **societal impacts of AI**.

Education

- 2021 – 2027 (*exp.*) **Cornell University**, Ithaca, NY, USA
PhD in Statistics
Committee chair: Martin T. Wells · Advisors: Lionel Levine, Moon Duchin
- 2021 – 2024 **Cornell University**, Ithaca, NY, USA
MS in Statistics GPA: 3.93 / 4.00
- 2019 – 2021 **Indian Statistical Institute**, Bengaluru, India
Master of Mathematics (M.Math)
First Division with Distinction
- 2016 – 2019 **Indian Statistical Institute**, Bengaluru, India
Bachelor of Mathematics (B.Math)
First Division with Distinction

Publications and Preprints

- 2026 **EigenBench: A Comparative Behavioral Measure of Value Alignment**
J. Chang, L. Piff, **Sana, S.**, J. X. Li, L. Levine
International Conference on Learning Representations (ICLR), 2026
Oral presentation (~ top 1% of submissions) · [Paper](#)
- 2026 **Pluralistic Preference Alignment via Sortition-Weighted RLHF**
Sana, S., J. Wu, M. T. Wells
ICML 2026 Workshop on Pluralistic Alignment · [arXiv:2602.05113](#)
- 2026 **Gradation of Arrow's Axioms**
Sana, S., P. Rock, M. T. Wells, M. Duchin
Submitted to the Annals of Applied Statistics · [Preprint](#)
- 2026 **Mixing Times of Glauber Dynamics on Masked Language Models**
Sana, S., S. Wolf, N. Mehta, A. Shah, A. Shaikh, J. Goodman, L. Levine
Submitted to NeurIPS 2026 · [arXiv:2605.16378](#)

2026 **Statistical Aspects of Sortition**
Sana, S., M. T. Wells · *In preparation*

2026 **AI Value Evolution**
Sana, S., L. Levine · *In preparation*

Technical Reports

2025 **Quantitative Relaxations of Arrow's Axioms**
Sana, S., D. Brous, M. T. Wells, M. Duchin
Preliminary version of *Gradation of Arrow's Axioms* · [arXiv:2506.12961](https://arxiv.org/abs/2506.12961)

2024 **A Scalar Matching Factor on the Birkhoff Polytope Characterizing Permutation and Uniform Matrices**
Sana, S. · [arXiv:2406.16284](https://arxiv.org/abs/2406.16284)

Research and Internship Experience

Fall 2026 **AI Engineering Intern**, New York City Council Data Team
Susan and Rubinstein PiTech Innovation Fund, Cornell Tech · Advisor: Alaa Moussawi
Optimizing the retrieval-augmented generation (RAG) system used by NYC Council staff for legislative research.

Summer 2026 **PiTech Impact Fellow (AI Engineering)**, New York City Council Data Team
Advisor: Alaa Moussawi
Built an AI routing system and MCP server powering a staff-facing chatbot that directs queries across multiple generative-AI applications and external tools, giving NYC Council staff faster, more accurate answers to complex legislative questions through a single interface.

Summer 2025 **Summer Fellow**, Data and Democracy Lab, Cornell Brooks School of Public Policy
Advisor: Moon Duchin
Developed statistical fairness measures for ranked-choice voting systems, enabling empirical evaluation of how closely electoral outcomes align with principles from social choice theory.

Apr – Jun 2020 **Statistical Analyst Intern**, Siemens Healthineers, Bengaluru, India

Awards and Honors

2026 Susan and Rubinstein Innovation Fund, Cornell Tech

2026 ICML 2026 Silver Reviewer Award

2026 Cornell Bowers CIS Social Impact and Service Award

2026 Cornell PiTech Impact Fellowship

2025 Tapia Conference Travel Award (\$1,500)

2024 NSF Travel Award (\$500, IISA 2024)

2024	Cornell Graduate School Research Travel Grant (\$1,300, <i>study of ranked-choice elections</i>)
2021 – 2022	Cornell PhD Graduate Fellowship (<i>awarded to two incoming PhD students</i>)
2020	NBHM Master’s Scholarship, Government of India
2019	All India Rank 3, M.Stat entrance examination, Indian Statistical Institute
2019	All India Rank 33, IIT JAM (Mathematical Statistics)
2016 – 2019	ISI Teachers’ Award, Indian Statistical Institute (<i>awarded to top 3 students</i>)

Invited Talks and Presentations

Feb 2026	Social Choice for AI Ethics and Safety Workshop, IASEAI 2026, Paris, France (<i>invited talk</i>)
Jan 2026	Alignment Research Center, Berkeley, CA, USA (<i>invited talk</i>)
Oct 2025	Social Choice: Theory and Computation, Wellesley College, MA, USA (<i>poster</i>)
Jul 2025	Fairness in Social Choice Workshop, Northwestern University, IL, USA (<i>invited talk</i>)
Jan 2025	Joint Mathematics Meetings, Seattle, WA, USA (<i>contributed talk</i>)
Dec 2024	IISA 2024, Kerala, India (<i>contributed talk</i>)
Sep 2024	Statistics Graduate Student Seminar, Cornell University (<i>invited talk</i>)
Aug 2022	Hyundai Vision Conference, Seoul, South Korea (<i>poster</i>)

Teaching

Spring 2026	Co-Instructor , MATH 1710: Statistical Theory and Application in the Real World, Cornell University
Summer 2025	Lecturer , Research Practices and Academic Culture Seminar, Cornell University
	Teaching Assistant, Cornell University
Spring 2026	MATH 2940: Linear Algebra for Engineers — weekly discussion sections and office hours
Fall 2023 – Fall 2025	MATH 6710: Probability Theory I (three semesters: Fall 2023, 2024, 2025) — office hours and grading
Spring 2025	MATH 6720: Probability Theory II — office hours and grading
Spring 2025	STSCI 4100: Multivariate Statistics — grading and lab sessions
Fall 2023	CS 6386: Data to Decisions — office hours
2023	STSCI 2100: Introductory Statistics (Winter and Summer sessions) — grading and office hours
Spring 2023	MATH 4720: Statistics — office hours and grading
Spring 2023	STSCI 5640: Statistics for Financial Engineering — grading
Fall 2022	STSCI 4090: Theory of Statistics — office hours and grading

Mentorship and Service

<i>Sep 2025 – Present</i>	Research Lead , Cornell AI Alignment Club
<i>Dec 2024</i>	Session Chair , IISA 2024
<i>2022</i>	Professional Development Chair , Statistics Graduate Society, Cornell University
<i>Spring 2022</i>	Graduate Ambassador , Cornell CIS DEI

Academic Reviewing

2026 – ICML 2026 · NeurIPS 2026 · ICML 2026 Pluralistic Alignment Workshop · AAAI 2027

Skills

Programming: Python, R (proficient); MATLAB (familiar)

Tools and frameworks: PyTorch, Hugging Face Transformers, RAG pipelines, MCP servers, FastAPI, Git, RunPod, VoteKit

Languages: Bengali (native), English, Hindi, Korean

Other Interests

Cooking, running, and traveling.